Imperial College London

Autoencoder Market Models for Interest Rates

Topics in Derivative Pricing - Coursework Report

Diane Murzi (CID: 06048769) Chedi Mnif (CID: 06064588) Hamza El Arji (CID: 06020926)

Supervisor: Dr. Vladimir V. Piterbarg

Abstract

The term structure of interest rates encodes market expectations of monetary policy, inflation, and credit conditions, yet compressing its dynamics into a small number of interpretable factors remains a central challenge in quantitative finance. Classical parametric approaches such as Nelson–Siegel are constrained by fixed functional forms that do not adapt to data and cannot exploit cross-currency regularisation.

We replicate and extend the Autoencoder Market Model of Sokol [1], training variational autoencoders on daily OIS swap rates for seven currencies over January 2023–January 2026 and benchmarking against Nelson–Siegel, extended NS, and Svensson specifications. The 2-dimensional conditional VAE outperforms the 3-factor Nelson–Siegel basis across all seven currencies (pooled Wilcoxon $p < 10^{-16}$; mean gain +3.01 bp), using one fewer latent dimension.

Beyond replication, we show that CVAE residuals generate economically meaningful trading signals: a 2s10s slope-residual strategy achieves an out-of-sample portfolio Sharpe of 3.99 versus −2.35 for the Nelson–Siegel benchmark, demonstrating that reconstruction accuracy does not imply signal quality. A curvature butterfly strategy yields Sharpe 5.12, carry decomposition confirms the alpha is genuine, and a cross-currency convergence strategy exploiting the shared latent geometry achieves Sharpe 4.57, with JPY pairs systematically underperforming non-JPY pairs (0.78 vs. 2.31), validating the economic interpretability of the latent space. These results establish deep latent-factor models as a credible alternative to classical curve parametrisations for both compression and relative-value signal generation.

Keywords: Variational Autoencoder, AEMM, OIS Swap Rates, Nelson–Siegel, Yield Curve Trading, Cross-Currency Relative Value.

1. Introduction and Motivation

Why the yield curve is hard to model

The yield curve, meaning the relationship between interest rates and maturity at a given point in time, is the cornerstone of fixed-income pricing, risk management, and monetary policy transmission. The yield curve is summarised by a few liquid rates but displays diverse shapes across economic regimes (upward, inverted, humped, or flat).

The challenge is to build a model that:

(i) compresses these shapes into a small number of state variables for tractability;

(ii) covers the realistic set of possible curve shapes without generating economically absurd ones;

(iii) generalises across currencies without requiring entirely separate models.

Classical approaches are dominated by the *Nelson–Siegel* (NS) parametrisation, which represents zero-coupon rates as a linear combination of three exponential factors (level (β_1), slope (β_2), and curvature (β_3)) with decay parameter λ, and its four-parameter Svensson extension. These models are widely used by central banks and practitioners for precisely calibrated purposes: they are interpretable, fast to calibrate (closed-form OLS for fixed λ), and their factors have intuitive economic meaning. Yet they suffer from two structural limitations that motivate the present work.

(i) First, the NS basis was *not designed to minimise reconstruction error*: it is an approximation arising from a Taylor expansion of a two-exponential forward rate model, and large swathes of the three-dimensional factor space $(\beta_1, \beta_2, \beta_3)$ correspond to curve shapes that have never been observed in practice

(ii) Second, NS factors are calibrated *independently per currency*, meaning there is no mechanism to exploit the well-documented cross-currency co-movement of yield curves or to regularise data-scarce markets using information from liquid ones.

The autoencoder approach

Sokol [1] proposes a radically different compression mechanism: train a variational autoencoder (VAE) jointly on multi-currency historical swap rates so that the latent space covers exactly the set of curve shapes that have actually occurred, and no more. The key geometric insight is that real yield curves live on a low-dimensional manifold embedded in $\mathbb{R}^7$ (the space of seven-tenor swap rates); the VAE learns this manifold directly from data, whereas NS assumes it to be parameterised by three exponential functions.

The appeal of the VAE approach is threefold:

(i) ***Parsimony with data adaptation***: because the encoder learns from thousands of real curves across thirteen currencies in the original paper, two latent dimensions suffice to achieve accuracy comparable to the three-factor NS basis.

(ii) ***Realism enforcement***: any point decoded from the latent space produces a historically plausible curve, because the decoder is trained only on observed shapes.

(iii) ***Cross-currency regularisation***: a single shared encoder learns universal curve geometry, enabling the model to transfer information from data-rich currencies (USD, EUR) to data-scarce ones.

2. Mathematical Framework

Yield curve representation

We work with $N = 7$ OIS par swap rates at tenors $\tau_n \in \{2, 3, 5, 10, 15, 20, 30\}$ years. On each business day t, the yield curve is a vector $\mathbf{S}(t) \in \mathbb{R}^7$ expressed in decimal form. Following Sokol [1], we fit all models directly to par swap rates rather than first stripping a zero-coupon curve, eliminating bootstrap-induced noise and subjective interpolation choices.

The Nelson-Siegel family

The zero-coupon rate in the Nelson-Siegel model is

$$R(\tau) = \beta_1 + \beta_2\,\phi(\lambda\tau) + \beta_3\big[\phi(\lambda\tau) - e^{-\lambda\tau}\big], \qquad (1)$$

where $\phi(x) = (1 - e^{-x})/x$ and $\lambda > 0$ controls the decay speed. The three factors have transparent economic interpretations: β_1 is the long-run level (since $R(\infty) = \beta_1$), β_2 governs the slope (since $R(0) = \beta_1 + \beta_2$ under the appropriate limit), and β_3 controls the curvature hump, which peaks at $\tau^\star \approx 1/\lambda$. Given λ, the model is linear in $(\beta_1, \beta_2, \beta_3)$ and fitted by closed-form OLS in a single pass, a computational advantage that NS retains over all neural alternatives.

Equation (1) was not derived by minimising reconstruction error: it is an approximation of a two-exponential forward rate model obtained by setting both exponential decay rates equal and applying a first-order Taylor expansion. This historical origin is important: it means the NS basis functions were chosen for mathematical convenience, not for empirical fitness. Sokol's central argument is that a data-driven basis, learnt by a VAE, will necessarily do better, and our results confirm this.

The Svensson extension augments the Nelson–Siegel specification with an additional curvature factor governed by a second decay parameter λ_2:

$$R^{\mathrm{NSS}}(\tau) = R^{\mathrm{NS}}(\tau) + \beta_4\big[\phi(\lambda_2\tau) - e^{-\lambda_2\tau}\big], \qquad (2)$$

allowing for more flexible shapes such as double-humped curves. We use NSS only as a flexible benchmark and upper bound on reconstruction performance, since it requires per-date nonlinear calibration and does not generalise naturally across currencies.

For the Nelson–Siegel benchmark, we do not reuse a prespecified decay parameter from the literature or from Sokol. Instead, we estimate a single global value $\lambda^\star$ by pooled cross-validation across all currencies and dates, selecting the value that minimises the mean RMSE. This pooling ensures a fair comparison with the multi-currency VAE, which is trained jointly across currencies. We obtain $\lambda^\star \approx 0.500$.

Variational Autoencoder

A VAE consists of a probabilistic encoder $q_\phi(\mathbf{z} \mid \mathbf{S})$ and a decoder $p_\theta(\mathbf{S} \mid \mathbf{z})$, both parameterised by neural networks. The encoder maps a N-dimensional curve to a Gaussian distribution over a K-dimensional latent space:

$$\mathcal{E}_\phi(\mathbf{S}) = \big(\boldsymbol{\mu}(\mathbf{S}),\ \log \boldsymbol{\sigma}^2(\mathbf{S})\big) \in \mathbb{R}^{2K}.$$

The decoder maps a latent sample back to reconstructed rates via a Sigmoid output that keeps values in $[0, 1]$ (after normalisation): $\mathcal{D}_\theta : \mathbb{R}^K \to [0, 1]^N$.

During training, a latent sample is drawn using the *reparameterisation trick* [6]:

$$z_k = \mu_k(\mathbf{S}) + \sigma_k(\mathbf{S})\,\omega_k, \quad \omega_k \sim \mathcal{N}(0, 1),$$

which makes the sampling step differentiable and allows gradients to flow through the stochastic layer. At inference time we use the deterministic posterior mean $\mathbf{z} = \boldsymbol{\mu}(\mathbf{S})$, which is the standard convention for encoding applications.

The training objective is the Evidence Lower Bound (ELBO) under a Gaussian prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I_K)$. Matching Equation (5) of Sokol [1] exactly:

$$\mathcal{L}_{\mathrm{VAE}} = \underbrace{\frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}}\sum_{n=1}^{N}\big(S'_{i,n} - S_{i,n}\big)^2}_{\text{reconstruction: sum over tenors, mean over batch}} + \beta\, D_{\mathrm{KL}}, \qquad (3)$$

where the KL divergence for diagonal Gaussians is

$$D_{\mathrm{KL}} = -\tfrac{1}{2}\sum_{k=1}^{K}\big(1 + \log\sigma_k^2 - \mu_k^2 - \sigma_k^2\big).$$

We set $\beta = 10^{-7}$ throughout, following the paper. This extremely small weight places essentially all pressure on reconstruction accuracy, using the KL term only as a "light touch" regulariser to prevent gaps in latent space. The consequence is that the trained latent codes do not conform to the $\mathcal{N}(0, I)$ prior, all currencies have $z_2 \in [1.34, 1.88]$ far from zero, which strongly suggests that the VAE is operating almost as a deterministic autoencoder. This is a deliberate modelling choice [1], not a failure of training.

A critical implementation detail, which a naive port of the formula misses: the reconstruction term must be *summed* over the $N = 7$ tenors per observation and then *averaged* over the batch. Averaging over both dimensions would divide the reconstruction term by 7, artificially downweighting it relative to the KLD term.

After training, the sampling noise can be removed via *post-processing gradient descent* [1]: starting from the encoder mean $\boldsymbol{\mu}(\mathbf{S})$, we run 100 Adam steps directly on $\mathbf{z}$ to minimise reconstruction loss only, recovering the true L2-optimal latent code. For the multi-currency VAE this improves mean RMSE from 5.49 to 5.35 bp (+2.5%).

Conditional VAE

The conditional VAE (CVAE) [7] conditions both encoder and decoder on a one-hot currency label $\mathbf{y} \in \{0,1\}^C$ with $C = 7$:

$$q_\phi(\mathbf{z} \mid \mathbf{S}, \mathbf{y}), \quad p_\theta(\mathbf{S} \mid \mathbf{z}, \mathbf{y}).$$

Concretely, $\mathbf{y}$ is concatenated to the encoder input ($7 + 7 = 14$ dimensions) and to the decoder input ($2 + 7 = 9$ dimensions), so the model receives explicit currency identity. This gives each currency a dedicated neighbourhood of the latent space, allowing the decoder to learn currency-specific fine adjustments (level, slope regime) without the shared latent code $\mathbf{z}$ being burdened by that task. The training loss is identical to (3) but conditioned on $\mathbf{y}$.

Two CVAE architectures exist in the literature. Sokol uses the Type 1 formulation of Sohn, Lee, and Yan [7], where $\mathbf{y}$ is an *input* to both encoder and decoder. This is the correct choice here: the currency is always known at inference time, and there is no value in asking the decoder to predict it. The Type 2 formulation (where $\mathbf{y}$ is an output of the decoder) would be ill-posed because the same curve shape can appear in multiple currencies.

Multi-currency VAE and the shared latent space

The unconditional multi-currency VAE drops the currency label entirely and trains a plain VAE on the combined dataset of all currencies. Every observation, regardless of currency, is encoded into the *same* 2D latent space. This creates a "world map" of yield curve shapes where currencies with structurally similar regimes cluster together, a property that the CVAE lacks because its conditional architecture partitions the latent space by currency.

For trading purposes, the shared latent space is the key asset of Extension 2. If two currencies that are normally close in latent distance suddenly drift apart, their curve shapes have diverged more than usual, signalling a relative-value opportunity. This cross-currency signal is impossible to construct from NS factors (which have no shared coordinate system) or from the CVAE (whose latent codes are currency-specific). Only the multi-currency VAE's shared encoder creates a valid and economically interpretable cross-currency distance metric.

3. Neural Network Architectures

We implement the three architectures from Tables 1–3 of Sokol [1], adapted to $C = 7$ currencies, all with latent dimension $K = 2$. The three models have 99, 191, and 433 trainable parameters respectively, deliberately shallow to prevent overfitting and to keep the compression interpretable. The CVAE concatenates the one-hot currency vector $\mathbf{y}$ to both encoder and decoder inputs ($7+7 = 14$ and $2+7 = 9$ dimensions), following the Type-1 formulation of [7]. All models are trained for 5,000 epochs with Adam ($\eta = 10^{-3}$), batch size 256, gradient clipping at max-norm 1.0, and a ReduceLROnPlateau scheduler (patience 500, factor 0.5, floor 10^{-6}).

Table 1. Single-currency VAE (99 parameters) .

Layer	In	Out	Activation
Encoder linear 1	7	4	Tanh
Encoder linear 2	4	4	—
Reparameterisation	4	2	$z = \mu + \sigma\omega$
Decoder linear 1	2	4	Tanh
Decoder linear 2	4	7	Sigmoid

Table 2. Multi-currency VAE (191 parameters).

Layer	In	Out	Activation
Encoder linear 1	7	7	Tanh
Encoder linear 2	7	4	—
Reparameterisation	4	2	$z = \mu + \sigma\omega$
Decoder linear 1	2	4	Tanh
Decoder linear 2	4	7	Tanh
Decoder linear 3	7	7	Sigmoid

Table 3. Multi-currency CVAE (433 parameters).

Layer	In	Out	Activation
Encoder linear 1	7+7	13	Tanh
Encoder linear 2	13	4	—
Reparameterisation	4	2	$z = \mu + \sigma\omega$
Decoder linear 1	2+7	7	Tanh
Decoder linear 2	7	7	Tanh
Decoder linear 3	7	7	Sigmoid

4. Data and Experimental Setup

Bloomberg OIS data

We use daily OIS swap rates for seven currencies (AUD, CAD, DKK, EUR, GBP, JPY, USD) at seven tenors (2, 3, 5, 10, 15, 20, 30 years), downloaded from the Bloomberg Terminal as a historical data spreadsheet. Rates are quoted in percent and converted to decimal form on loading. The window 30 January 2023 to 15 January 2026 yields **774 aligned business days** per currency, for a pooled panel of $774 \times 7 = 5{,}418$ observations.

CHF was initially included but excluded after identifying that the Bloomberg ticker available to us predates the SARON migration. The series quoted CHF 2Y rates of 3–6% in 2023–2025, inconsistent with the true CHF SARON 2Y level of 0.5–1.8%. Including a misspecified series would contaminate the multi-currency VAE's

Table 4. Dataset comparison.

	This report	Sokol (2022)
Currencies	7	13
Rate type	OIS only	LIBOR + OIS
Period	2023–2026	∼2000–2022
Frequency	Daily	Monthly
Obs/currency	774	∼250
Tenors	7 (2–30Y)	7 (2–30Y)

shared latent space; the exclusion is a deliberate quality choice.

Sample characteristics

The sample is deliberately short but rich in regimes. AUD, CAD, GBP and USD experienced 2s10s inversions of −100 to −150 bp during the 2023 hiking cycle, a severity not seen since the early 1990s. EUR and DKK transitioned from inverted to positively sloped during 2024–2025 as the ECB began cutting rates. JPY remained structurally different from all other currencies: rates were confined to 0.2 - 1.5% across all tenors throughout the sample period, reflecting the Bank of Japan's yield-curve control policy (formally lifted in March 2024 but with persistent effects on curve shape through 2025). This diversity makes the dataset a demanding test for any multi-currency compression model.

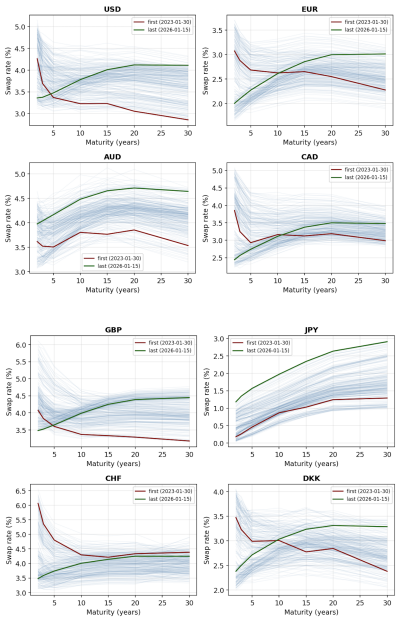


Figure 1. Historical OIS swap rate curves per currency. Each faint line is one observation; the first and last curves are highlighted in dark red and dark green. JPY's distinctive low-rate regime and the aggressive inversions of AUD, USD, and CAD are immediately visible.

Train/test split

We split at **31 March 2025** (approximately 70/30 chronologically), yielding 566 train dates and **208 OOS business days** (April 2025 to January 2026). NS fits are not split because NS is a per-date OLS problem, not a trained model with parameters to hold out. All trading backtests are evaluated strictly on the OOS window.

5. Nelson-Siegel Benchmark Results

Cross-validation of λ

Before benchmarking the VAE, we must give NS its best possible chance. Rather than adopting an arbitrary value we select $\lambda^\star$ by pooled cross-validation across all currencies and dates, minimising mean RMSE on a 40-point grid. This matters: a poorly chosen λ would handicap NS and make the VAE comparison unfair.

The minimiser is $\lambda^\star = 0.500$, placing the curvature hump at $\tau^\star = 2$ years. This is economically meaningful: our sample is dominated by the 2023–2024 inversion cycle, where the most active curve deformations occurred at the short end rather than at the classical 2.5–3 year belly. NS is already telling us, through its own optimal parameterisation, that the short end mattered most during our period.

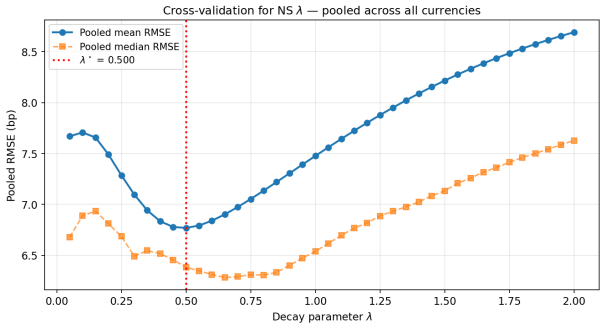


Figure 2. Pooled cross-validation of λ. The mean RMSE curve (blue) is unimodal with a clear interior minimum at $\lambda^\star = 0.500$ (red dotted), consistent with the short-end dominance of the 2023–2024 inversion cycle. The flatness around the minimum confirms that the NS benchmark is not sensitive to the precise choice of λ.

Per-currency and pooled RMSE

Table 5 reveals three salient patterns. First, the NS–NSS gap is 4.5× at the pooled level (6.77 vs. 1.52 bp), confirming that a single curvature term is systematically insufficient for the diverse curve shapes in our sample. Second, AUD has the worst NS RMSE (10.00 bp), driven by a systematic 30Y miss of 17.6 bp mean absolute residual; the AUD curve has a long-end that does not conform to the NS shape even at the optimal λ. Third, the JPY NS-to-eNS gain is dramatic: 6.49 → 2.59 bp (−60%). This is the largest gap of any currency and means that the global $\lambda^\star$ is very poorly adapted to JPY's near-zero, nearly flat rate structure. The VAE avoids this problem entirely because it imposes no parametric form: JPY simply occupies its own dedicated region of the shared latent space.

6. VAE Replication Results

The central finding: 2D beats 3D

The results in Table 6 confirm Sokol's thesis on our dataset. The 2D CVAE achieves 3.80 bp pooled RMSE

Table 5. Mean reconstruction RMSE (bp) by currency and model.

Model	AUD	CAD	DKK	EUR	GBP	JPY	USD	Pooled
NS (3D)	10.00	7.43	8.31	6.35	2.53	6.49	6.28	6.77
eNS (3D)	8.90	6.77	7.73	5.82	2.36	2.59	5.21	5.62
NSS (4D)	2.03	2.25	2.04	1.02	0.45	1.53	1.32	1.52

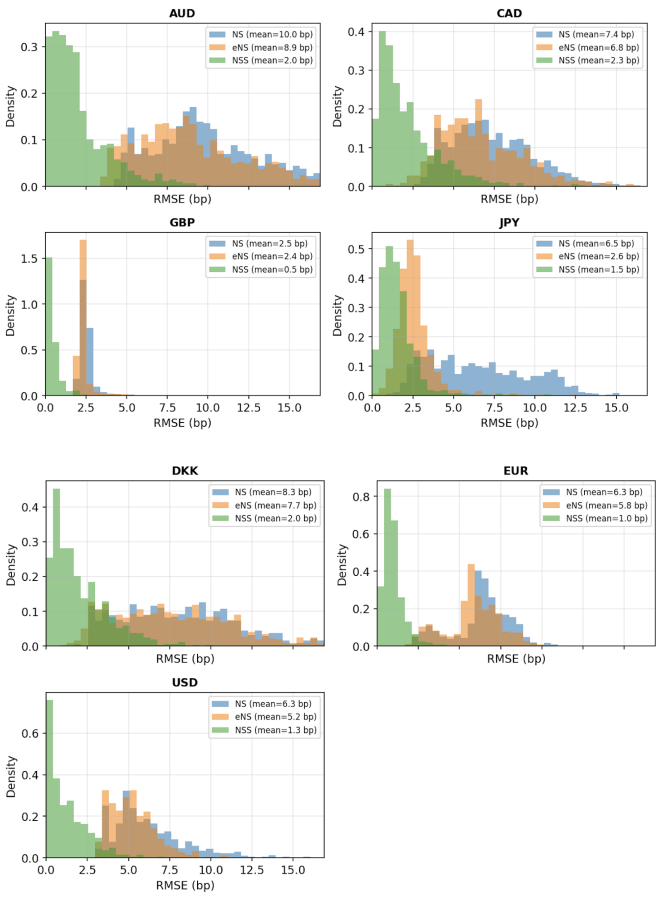


Figure 3. Per-currency in-sample RMSE distributions for NS, eNS, and NSS (replication of Sokol Fig. 11). The hierarchy NSS < eNS < NS holds without exception. JPY shows the largest NS–eNS gap; GBP the smallest (its regular curve shape is exceptionally well-fitted by the 3-parameter exponential basis

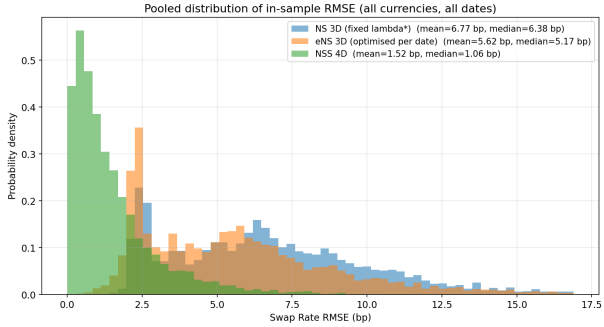


Figure 4. Pooled RMSE distributions across all currencies and dates. NS mean: 6.77 bp. NSS: 1.52 bp. The target for the 2D VAE is to close this gap without the extra dimension that NSS requires.

Table 6. Mean reconstruction RMSE (bp) by model and currency. Bold values mark the best performer per currency. The 2D CVAE outperforms the 3D NS basis on all seven currencies

Model	AUD	CAD	DKK	EUR	GBP	JPY	USD	Pooled
NS (3D)	10.00	7.43	8.31	6.35	2.53	6.49	6.28	6.77
NSS (4D)	2.03	2.25	2.04	1.02	0.45	1.53	1.32	1.52
Single-VAE (2D)	12.97	11.85	15.75	13.20	2.48	4.99	8.11	9.90
Multi-VAE (2D)	5.77	6.10	6.28	3.38	6.78	4.63	5.52	5.49
CVAE (2D)	**4.00**	**4.53**	**4.88**	**2.75**	**2.49**	**3.00**	**4.64**	**3.80**

against 6.77 bp for the 3D NS basis, a +3.0 bp improvement using *one fewer latent dimension*. The four-dimension NSS (1.52 bp) remains more accurate: at $K = 2$, the CVAE's expressiveness is still limited, and the NSS's second curvature term addresses exactly the long-end distortions that the CVAE cannot fully resolve. The claim is narrower but sharper: *holding K fixed at the aggressively low value of 2, multi-currency conditioning plus a learnt encoder recovers more curve information than three independent linear coefficients.*

Posterior-collapse-like behaviour in the single-currency VAE

Several single-currency VAEs (AUD, CAD, DKK, EUR, USD) exhibit *posterior-collapse-like behaviour*: the encoder maps many inputs to very similar Gaussian distributions, and the decoder correspondingly produces near-mean curves with limited sensitivity to the input, yielding RMSE above NS (AUD: 12.97 vs. 10.00 bp). This does not appear to be a software bug, but rather a capacity mismatch: the 2D bottleneck combined with the normalisation interval $[-5\%, 25\%]$ constrains the decoder's dynamic range for currencies occupying only 12–18% of $[0, 1]$. Joint multi-currency training eliminates this behaviour almost completely: the CVAE achieves sub-5 bp on every currency. For practitioners, the lesson is practical rather than semantic: small VAE architectures at aggressive compression ratios require multi-currency joint training or a longer history spanning multiple regimes.

Multi-VAE underperformance on GBP

The unconditional multi-currency VAE beats NS on 6 of 7 currencies but trails it on GBP (6.78 vs. 2.53 bp). This reflects the inherent shared-latent-space trade-off: the decoder must bridge the large gap between JPY ($z_2 \approx 1.34$, ultra-low rates) and the Western cluster ($z_2 \approx 1.6$–1.9, 4–5% rates), sacrificing GBP accuracy, whose simple, regular curves are the easiest to fit, to better handle more complex currencies. The CVAE resolves this entirely via currency conditioning: GBP CVAE RMSE is 2.49 bp, below NS.

Statistical significance

The Wilcoxon signed-rank test on paired daily RMSE differences (non-parametric, robust to right-skewed distributions) rejects the null of zero median improvement with overwhelming statistical significance on all seven currencies individually and in the pooled test ($p < 10^{-16}$). The multi-VAE similarly beats NS on 6/7

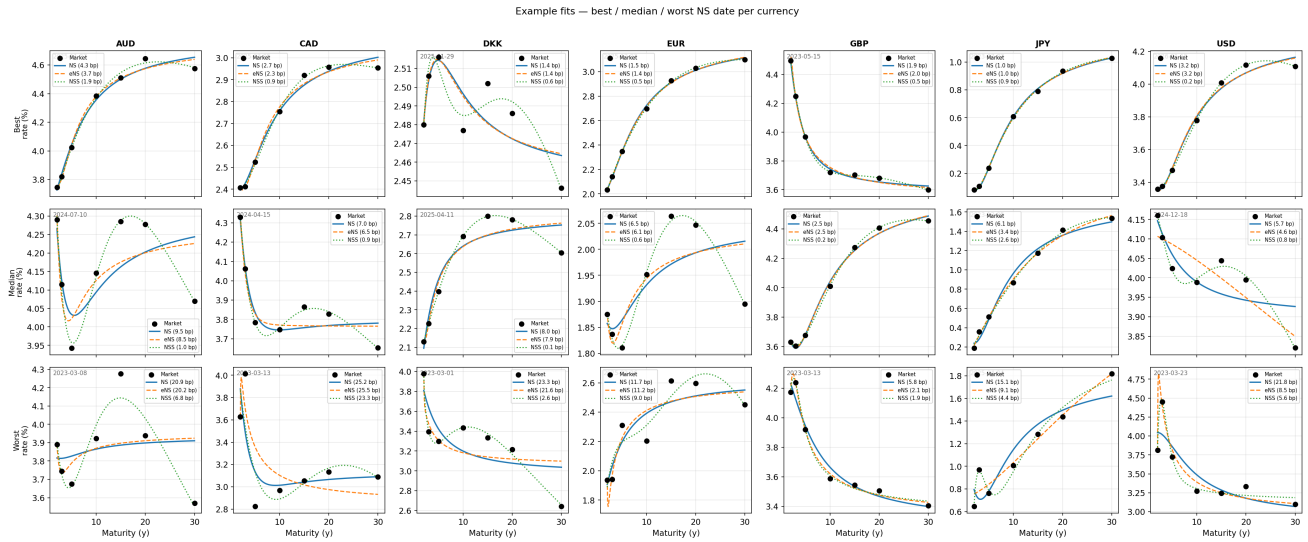


Figure 5. NS, eNS, and NSS fits on the best, median, and worst RMSE dates for each currency. On worst-case dates, curves exhibit double humps or kinks that the three-parameter NS cannot reproduce; the four-parameter NSS captures them by exploiting its second curvature term.

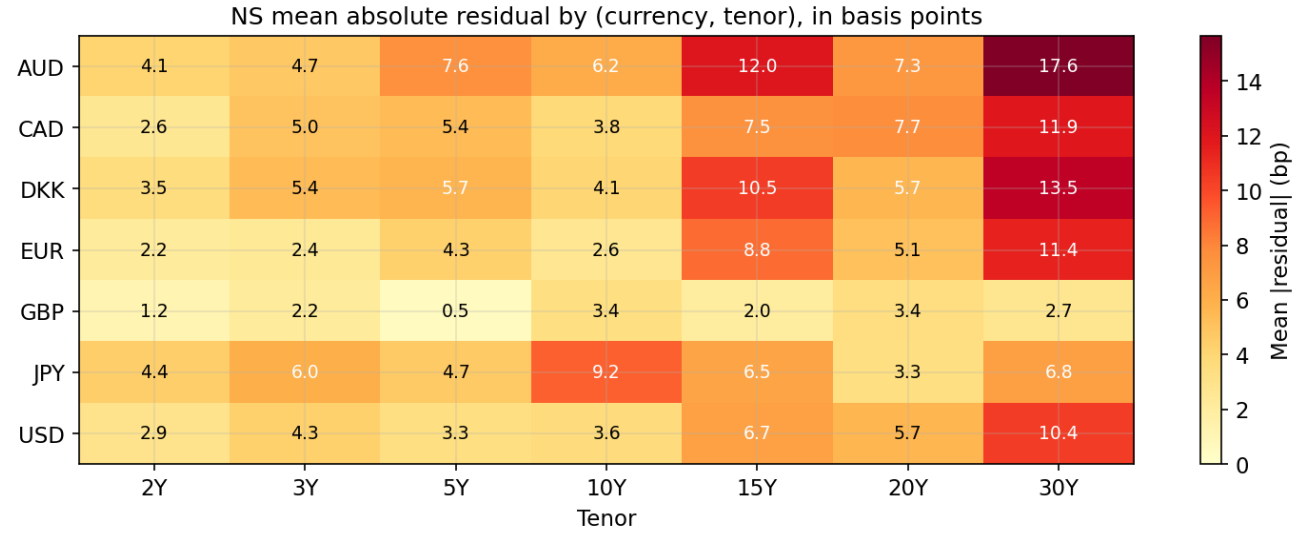


Figure 6. Mean absolute NS residual (bp) by currency and tenor. The long-end (20Y, 30Y) and the 15Y belly are the most problematic. The worst single cell is AUD 30Y at 17.6 bp. JPY 10Y at 9.2 bp reflects the BoJ's yield-curve control policy, which distorted the 10Y maturity independently of the rest of the curve. GBP residuals are uniformly small, confirming that GBP is the most NS-compatible currency in our sample.

Table 7. Wilcoxon signed-rank test: CVAE vs NS, paired on 774 daily RMSE differences per currency. Pooled test stacks all 5,418 observations.

Currency	$\overline{\Delta}$ (bp)	p-value	Winner
AUD	6.00	2.96×10^{-128}	CVAE
CAD	2.89	9.87×10^{-112}	CVAE
DKK	3.43	4.83×10^{-108}	CVAE
EUR	3.59	7.59×10^{-126}	CVAE
GBP	0.04	1.07×10^{-5}	CVAE
JPY	3.49	8.40×10^{-109}	CVAE
USD	1.63	7.59×10^{-83}	CVAE
Pooled	3.01	$< 10^{-16}$	**CVAE**

currencies (pooled $\overline{\Delta} = +1.28$ bp, $p = 2.64 \times 10^{-154}$), with GBP as the sole exception.

The Wilcoxon results should be interpreted cautiously because daily RMSE differences are serially dependent; the test is used here as descriptive evidence of systematic dominance rather than as a fully robust time-series inference procedure.

Latent space geometry

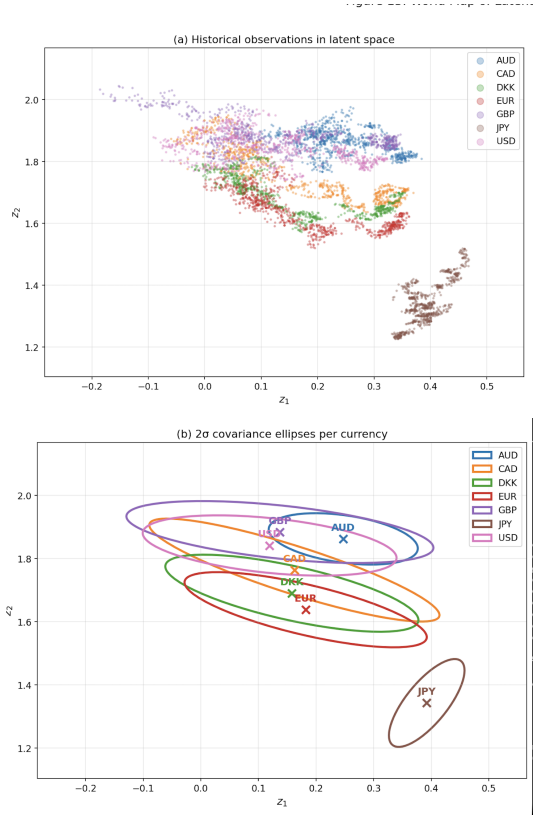


Figure 7. The multi-VAE latent space "world map" (replication of Sokol Fig. 13). Up: historical codes by currency. Bottom: 2σ covariance ellipses. JPY ($z_1 \approx 0.39$, $z_2 \approx 1.34$) is structurally isolated from all other currencies. EUR and DKK ellipses overlap almost perfectly, reflecting the EUR-DKK peg. CAD and USD partially overlap, reflecting Fed-BoC policy synchronisation.

The world map in Fig. 7 provides the most compelling visual evidence of the AEMM's value. The multi-VAE, without being told which currency each observation belongs to, spontaneously organises the latent space along economic lines. JPY occupies a structurally isolated island ($z_2 = 1.34$) reflecting the BoJ's decade-long near-zero rate policy. The six Western currencies form a compact cluster ($z_2 \in [1.6, 1.9]$) reflecting their synchronised post-COVID tightening and easing cycles. Within the cluster, EUR and DKK overlap (z_2 standard deviations of 0.059 and 0.061), consistent with the EUR-DKK peg; AUD and GBP show wider spreads in z_1 (0.065 and 0.133 respectively), reflecting greater curve-shape diversity over the period. This emergent economic ordering, discovered without any explicit supervision, is the clearest qualitative evidence for the VAE approach.

The latent variables carry clear economic interpretations: z_2 encodes the *rate level* (JPY at 1.34, AUD/GBP at ≈ 1.88), while z_1 encodes the *curve shape* (slope and curvature, with CAD showing the largest variation in z_1, consistent with its frequent inversion/normalisation transitions). This two-factor decomposition mirrors the classical level/slope interpretation of the Nelson-Siegel factors, but it is learnt from data rather than imposed by assumption.

Figure 14: Historical vs Reconstructed Swap Rates

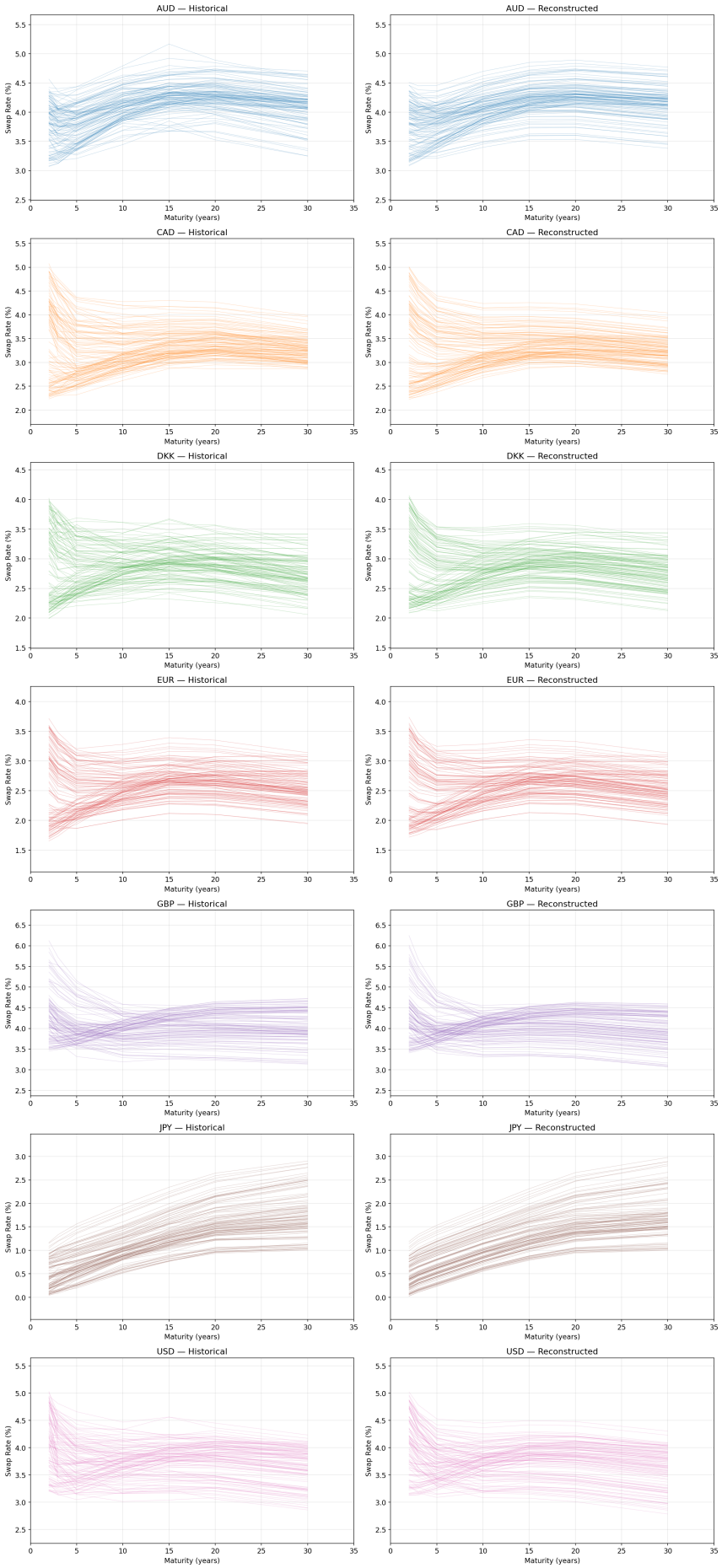


Figure 8. Historical (left) vs multi-VAE-reconstructed (right) swap rate curves per currency. The decoder faithfully reproduces the full range of curve shapes for each currency. The main visible effect of VAE regularisation is mild compression of extreme outliers toward the historical mean, a direct consequence of the KLD term even at $\beta = 10^{-7}$.

Figure 15: Curves Generated by Decoding Points on 2σ Ellipse Perimeter

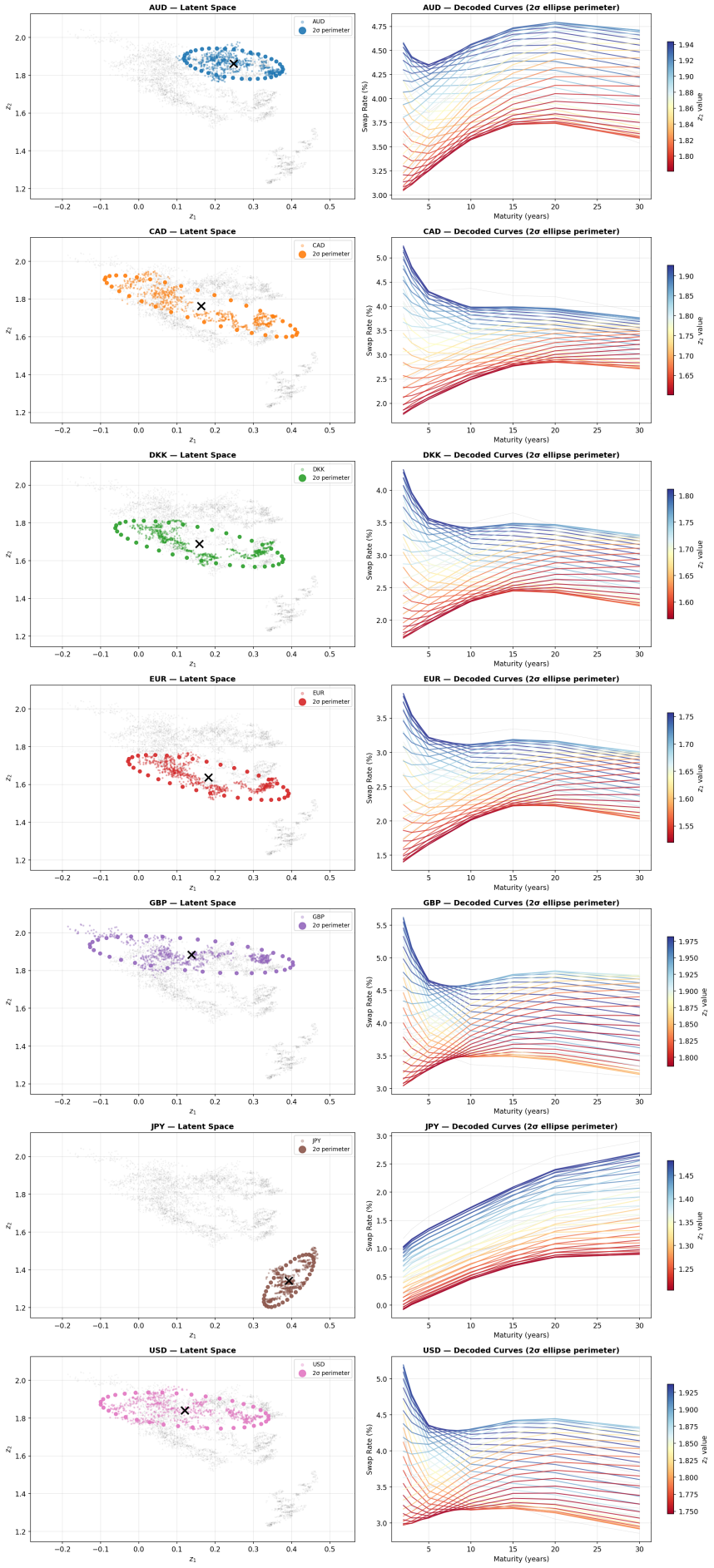


Figure 9. Curves decoded from the 2σ ellipse perimeter of each currency's latent distribution (replication of Sokol Fig. 15). Colour encodes z_2. For AUD, varying z_2 shifts the rate level; varying z_1 rotates the slope. For JPY, the entire perimeter corresponds to $z_2 \in [1.25, 1.45]$ and rates in $[0, 2.5\%]$—structurally disjoint from all other currencies.

7. AEMM Model Variants

The trained VAE encoder and decoder serve as building blocks for three interest-rate model classes from Chapters 3–4 of Sokol [1]. Each replaces the classical state variables (short rate, NS factors, forward rates) with the 2D latent code $\mathbf{z}_t = (z_{1,t}, z_{2,t})^\top$, reconstructing the full curve via the decoder at each step.

Forward Rate AEMM

The forward rate AEMM instantiates HJM dynamics in latent space: the latent code evolves as a diffusion process, and the resulting curve is decoded at each time step. With no drift constraint in the latent space, the model relies on re-encoding the decoded curve to maintain it on the learned manifold. We simulate 100 paths over 12 monthly steps with isotropic volatility $\boldsymbol{\sigma} = (0.01, 0.01)$.

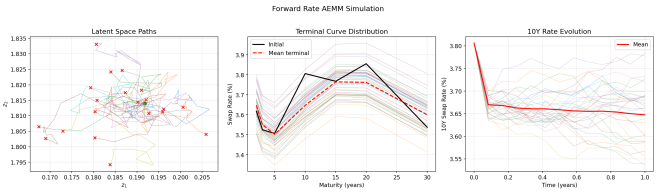


Figure 10. Forward Rate AEMM simulation (AUD). Left: latent space paths (green dots = start, red crosses = end). Centre: terminal curve distribution after 12 monthly steps; the re-encoding step keeps all curves within the historical manifold. Right: 10Y rate time evolution showing realistic diffusion.

Multi-factor short rate AEMM

The short rate AEMM assigns Ornstein-Uhlenbeck dynamics to each latent factor:

$$dz_k = \kappa_k(\theta_k - z_{k,t})\,dt + \sigma_k\,dW_{k,t}, \tag{4}$$

with $\boldsymbol{\kappa} = (0.1, 0.3)$ and $\boldsymbol{\sigma} = (0.005, 0.01)$. This is the AEMM equivalent of a two-factor Hull-White model, but with factors learnt from data rather than specified by parametric ansatz. The short rate is approximated by linear extrapolation of the decoded curve to zero maturity.

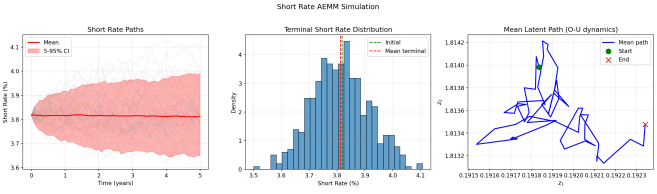


Figure 11. Multi-factor short rate AEMM (500 paths, 60 monthly steps). Left: short rate fan chart with 5–95% CI (red shading) and mean path (red line). Centre: terminal short rate distribution, symmetric and approximately Gaussian, as expected for Gaussian O-U dynamics. Right: mean latent path under O-U dynamics, converging back toward the mean level.

Autoregressive P-measure AEMM

The P-measure AEMM fits a first-order vector autoregression on the latent factors:

$$\mathbf{z}_{t+1} = \mathbf{c} + \mathbf{A}\,\mathbf{z}_t + \boldsymbol{\varepsilon}_{t+1}, \quad \boldsymbol{\varepsilon}_{t+1} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}). \tag{5}$$

Fitted on 774 observations: intercept $\mathbf{c} = (0.043, 0.009)^\top$, eigenvalues of $\mathbf{A}$ equal $0.9896 \pm 0.0053j$ with modulus $0.990 < 1$ (stationary). The near-unit modulus reflects the well-known high persistence of interest rate factors. This model is the 2D analogue of the Dynamic Nelson-Siegel VAR, but uses learnt latent factors instead of the parametric NS factors, reducing the VAR(1) parameter count from 15 to 7 and dramatically improving matrix conditioning.

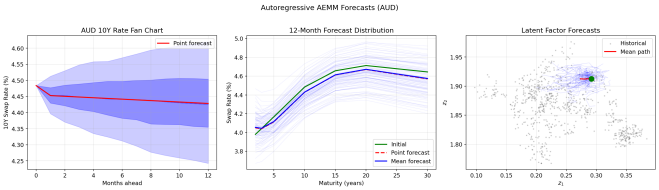


Figure 12. Autoregressive P-measure AEMM (AUD, 1,000 paths, 12-month horizon). Left: fan chart for the 10Y rate with three percentile bands. Centre: terminal curve distribution; the decoded curves are all economically plausible, confirming the manifold constraint. Right: latent factor forecasts in (z_1, z_2) space, radiating from the current AUD latent position.

8. Strategy A: CVAE Slope Residual (2s10s)

Economic rationale and signal construction

The CVAE produces a data-driven "fair value" for every swap curve by reconstructing it from the 2D latent code. When the actual 2s10s slope $(S_{10Y} - S_{2Y})$ deviates from the CVAE fair-value slope, this deviation cannot be explained by the learnt curve dynamics and is therefore likely idiosyncratic, transient noise. Mean reversion of such deviations is the classic relative-value trade.

Formally, the strategy signal is the CVAE slope residual:

$$\Delta_t^{\text{slope}} = \underbrace{(S_{10Y,t} - S_{2Y,t})}_{\text{actual slope}} - \underbrace{(S'_{10Y,t} - S'_{2Y,t})}_{\text{CVAE fair value}}, \tag{6}$$

where $\mathbf{S}'_t = \mathcal{D}_\theta(\boldsymbol{\mu}(\mathbf{S}_t, \mathbf{y}), \mathbf{y})$. This residual is standardised by a 60-day rolling z-score. The position rule applies hysteresis mean-reversion: enter a *flattener* when $z > +1.0$ (fade excess steepness), a *steepener* when $z < -1.0$ (fade excess flatness), and exit when $|z| < 0.3$. Positions are lagged by one bar to simulate execution at the following day's open.

We benchmark against two alternatives on the same OOS window: NS_resid (the identical logic applied to the NS fair-value residual using $\lambda^\star = 0.500$), and Naive (z-score of the raw slope with no model). P&L is reported in raw basis points (pos × daily slope change, with 0.5 bp transaction cost per flip) and also DV01-weighted (using the annuity approximation $\text{DV01}(T, y) \approx [1-(1+y)^{-T}]/y \cdot 10^{-4}$).

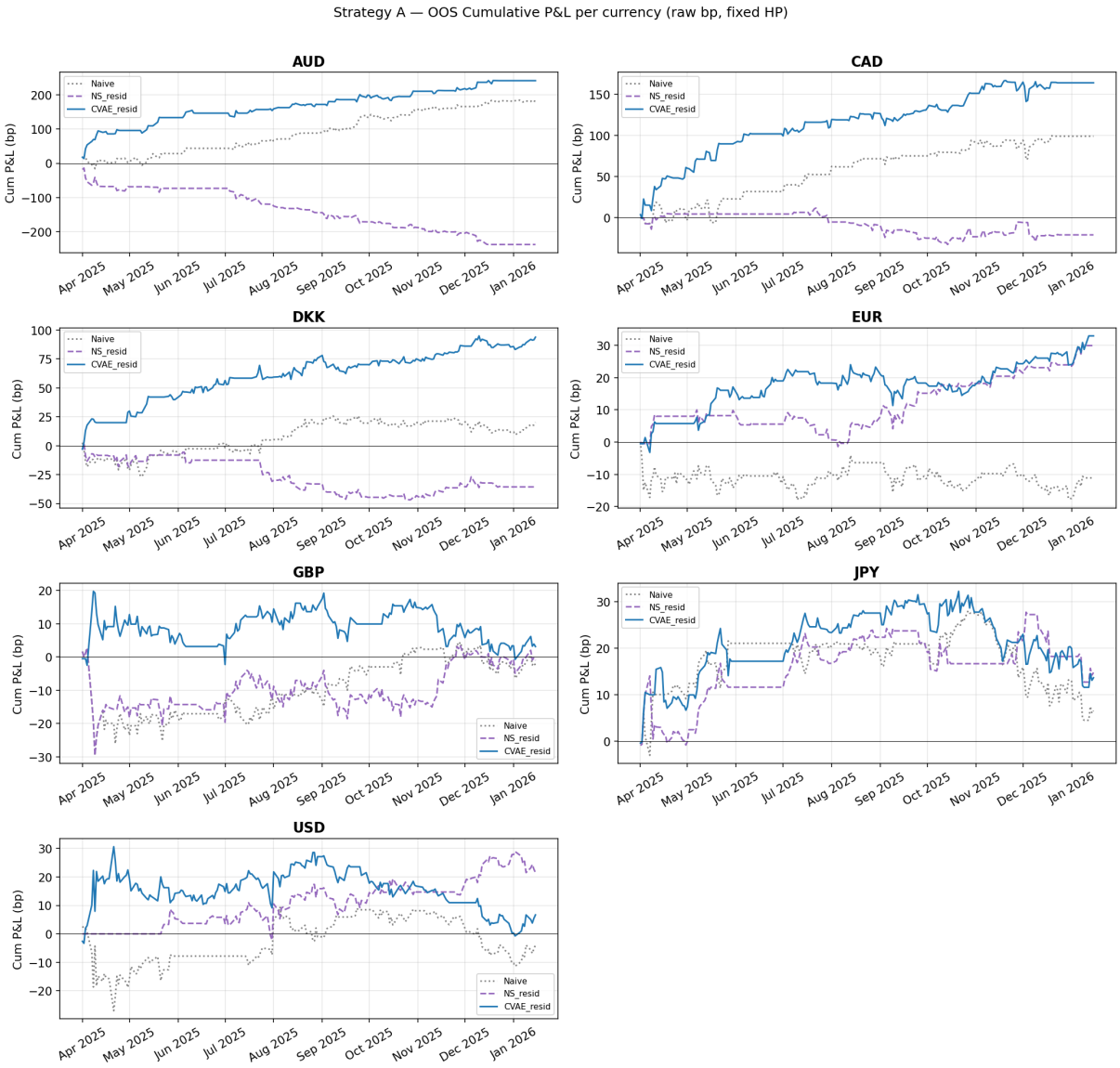


Figure 13. OOS cumulative P&L per currency for Strategy A (raw bp, fixed HP). CVAE_resid (blue solid) dominates on AUD (+3.77), CAD (+2.79), and DKK (+2.72). NS_resid (purple dashed) is negative on 4 of 7 currencies, including a Sharpe of −3.63 for AUD—the strategy actively destroys value by using the NS residual as a signal.

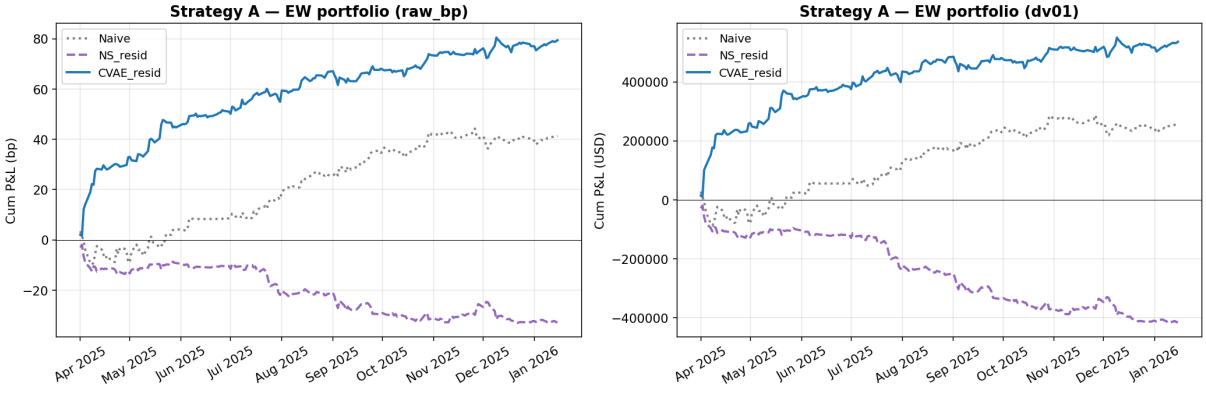


Figure 14. Equal-weight portfolio cumulative P&L, Strategy A. Left: raw bp. Right: DV01-weighted (USD notional basis). The CVAE_resid portfolio (Sharpe 3.99) diverges from the NS_resid portfolio (Sharpe −2.35) immediately and consistently across the full OOS window.

Table 8. Per-currency OOS Sharpe ratio (raw bp, fixed HP) and equal-weight portfolio.

Currency	Naive	NS_resid	CVAE_resid
AUD	2.83	−3.63	**+3.77**
CAD	1.78	−0.54	**+2.79**
DKK	0.52	−1.11	**+2.72**
EUR	−0.45	1.85	**+1.69**
GBP	−0.08	−0.03	**+0.10**
JPY	0.28	0.71	**+0.58**
USD	−0.11	0.95	**+0.18**
Portfolio EW	2.18	−2.35	**+3.99**

Table 9. Equal-weight OOS portfolio statistics, Strategy A (raw bp and DV01-weighted). The Sharpe ratio, annualised P&L, and maximum drawdown are all reported.

Signal	Sharpe	Total (bp)	Ann (bp)	MaxDD (bp)	HitRate
Raw basis-point P&L					
CVAE_resid	3.99	79.4	96.2	−5.9	0.56
Naive	2.18	41.3	50.1	−13.3	0.54
NS_resid	−2.35	−32.8	−24.8	−31.1	0.46
DV01-weighted P&L (USD 10mm notional)					
CVAE_resid	3.22	537 k	650 k	−52 k	0.54
Naive	1.60	255 k	309 k	−115 k	0.53
NS_resid	−3.53	−293 k	−354 k	−543 k	0.42

The central trading finding

The CVAE_resid portfolio Sharpe of **3.99** versus −2.35 for NS_resid is the most economically significant result in this report. Nelson-Siegel achieves a pooled in-sample RMSE of 6.77 bp, so it fits the curve well, yet its residuals produce a portfolio Sharpe of −2.35 when used as a trading signal. This demonstrates, with stark clarity, that a good *fitting* model is not automatically a good *signal* generator.

The explanation is mechanical: on a short 208-day window, the three-parameter NS OLS fit changes rapidly as the curve shape evolves across the hiking and easing cycle, producing high-frequency sign changes in the residual. Any strategy that trades mean-reversion on this residual is whipsawed by the noise, generating excessive turnover and net negative P&L after transaction costs. The CVAE's multi-currency pooling acts as a natural smoother: shared encoder weights regularise the residual to reflect genuine structural deviations rather than per-date fitting noise.

Hyperparameter robustness

To confirm that the CVAE signal is not the result of overfitting to a particular parameter configuration, we grid-search the entry threshold $\in \{0.75, 1.0, 1.25, 1.5\}$, exit threshold $\in \{0, 0.2, 0.4\}$, and lookback $\in \{30, 60, 90\}$ days on the *training window only*, then evaluate the best configuration out-of-sample. The fixed-HP portfolio Sharpe is 3.99; the tuned-HP portfolio Sharpe is also 3.99 ($\Delta = 0.00$). A spurious signal would show a large positive Δ from in-sample tuning. The absence of any improvement is strong evidence that the signal is genuine.

9. Strategy B: CVAE Curvature Residual (2s5s10s Butterfly)

Why curvature and why the butterfly

The 2s5s10s butterfly

$$b_t = 2 \cdot S_{5Y,t} - S_{2Y,t} - S_{10Y,t} \tag{7}$$

captures the departure of the 5Y rate from the linear interpolation between 2Y and 10Y. By construction it is DV01-neutral to both parallel shifts and slope changes, isolating pure curvature. Bikbov and Chernov [5] showed that curvature is the most predictable of the three NS principal components; the butterfly is the natural instrument for trading that predictability.

The CVAE fair-value butterfly $b'_t = 2 \cdot S'_{5Y,t} - S'_{2Y,t} - S'_{10Y,t}$ provides a data-driven curvature benchmark. The nonlinear decoder has particular advantages here: the 5Y point is where the NS exponential basis struggles most (the belly of the hump that NS cannot fully parameterise), so the CVAE's learned basis produces cleaner 5Y residuals. The butterfly is also approximately carry-neutral (the coupon accruals of the three legs partially cancel), so its P&L is attributable to pure curvature alpha rather than a carry trade in disguise, verified explicitly in Extension 1.

Table 10. Per-currency and portfolio OOS Sharpe, Strategy B (raw bp).

Currency	Naive_bf	NS_bf	CVAE_bf
AUD	3.83	2.64	**+3.09**
CAD	2.83	0.65	**+2.05**
DKK	2.61	0.91	**+3.03**
EUR	3.71	3.76	**+3.99**
GBP	−1.57	−1.57	−1.60
JPY	0.76	−1.47	0.63
USD	−0.94	−2.01	−1.24
Portfolio EW	5.39	3.36	**+5.12**

Strategy B achieves a portfolio Sharpe of **5.12**, demonstrating the superior fit quality of the CVAE at the curvature dimension relative to NS (3.36). We note honestly that the Naive butterfly Sharpe (5.39) is marginally above the CVAE (5.12). The explanation is structural: during the 2025 OOS window, the post-COVID curve normalisation made butterflies strongly mean-reverting around their historical averages, so the naive z-score of the raw butterfly captured this movement without requiring a model. The CVAE adds substantial value over NS (3.36 → 5.12) but does not add incremental value over the naive signal on this specific 9-month window. This honest result underscores the importance of a longer sample for fully discriminating between model-driven and naive signals.

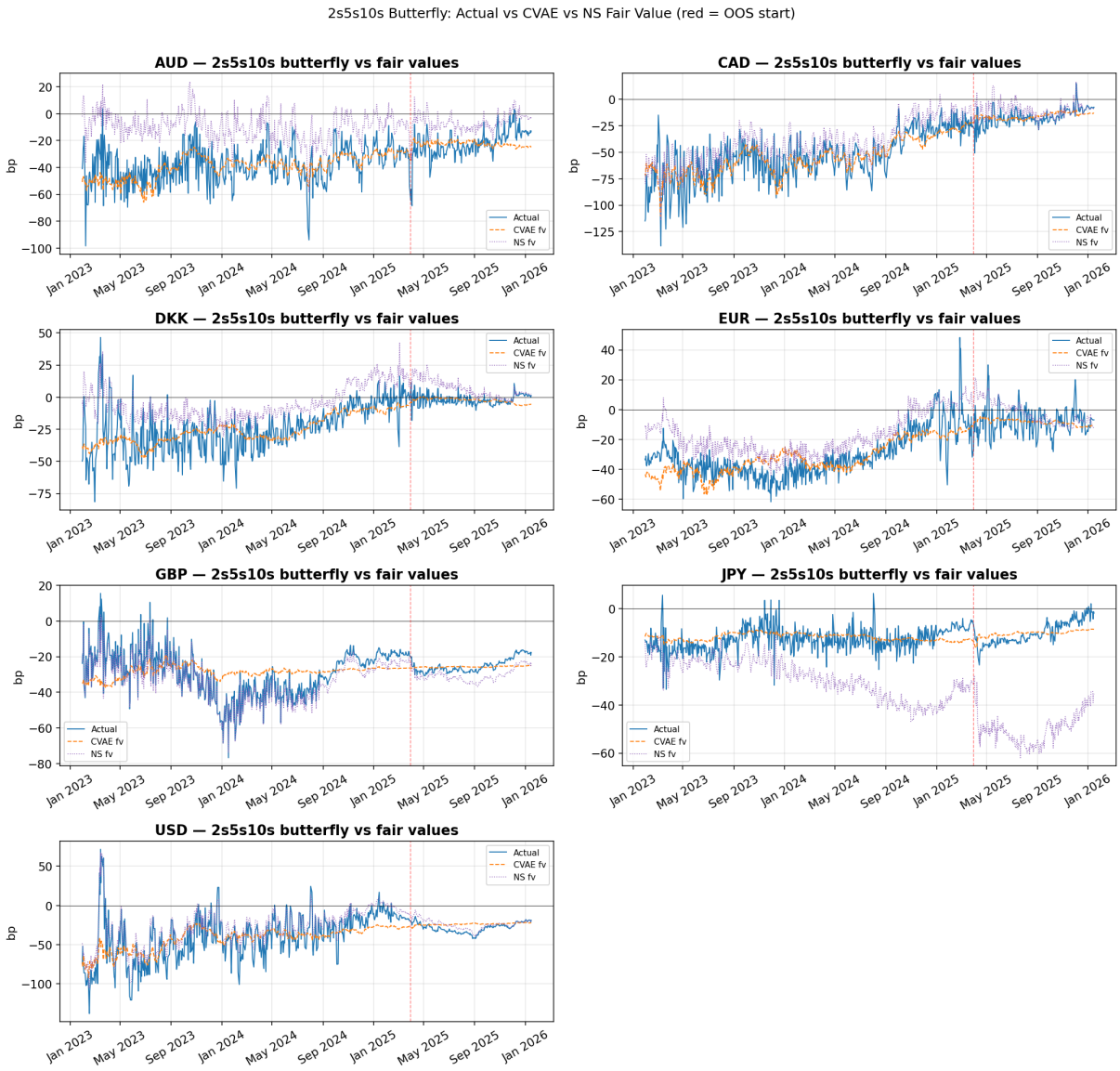


Figure 15. Actual vs CVAE vs NS fair-value butterfly for each currency. The red dashed line marks the train/OOS cutoff (31 March 2025). The CVAE fair value (orange dashed) tracks the actual butterfly more closely than the NS fair value (purple dotted) across all seven currencies, particularly at the USD belly where the NS exponential basis is most constrained.

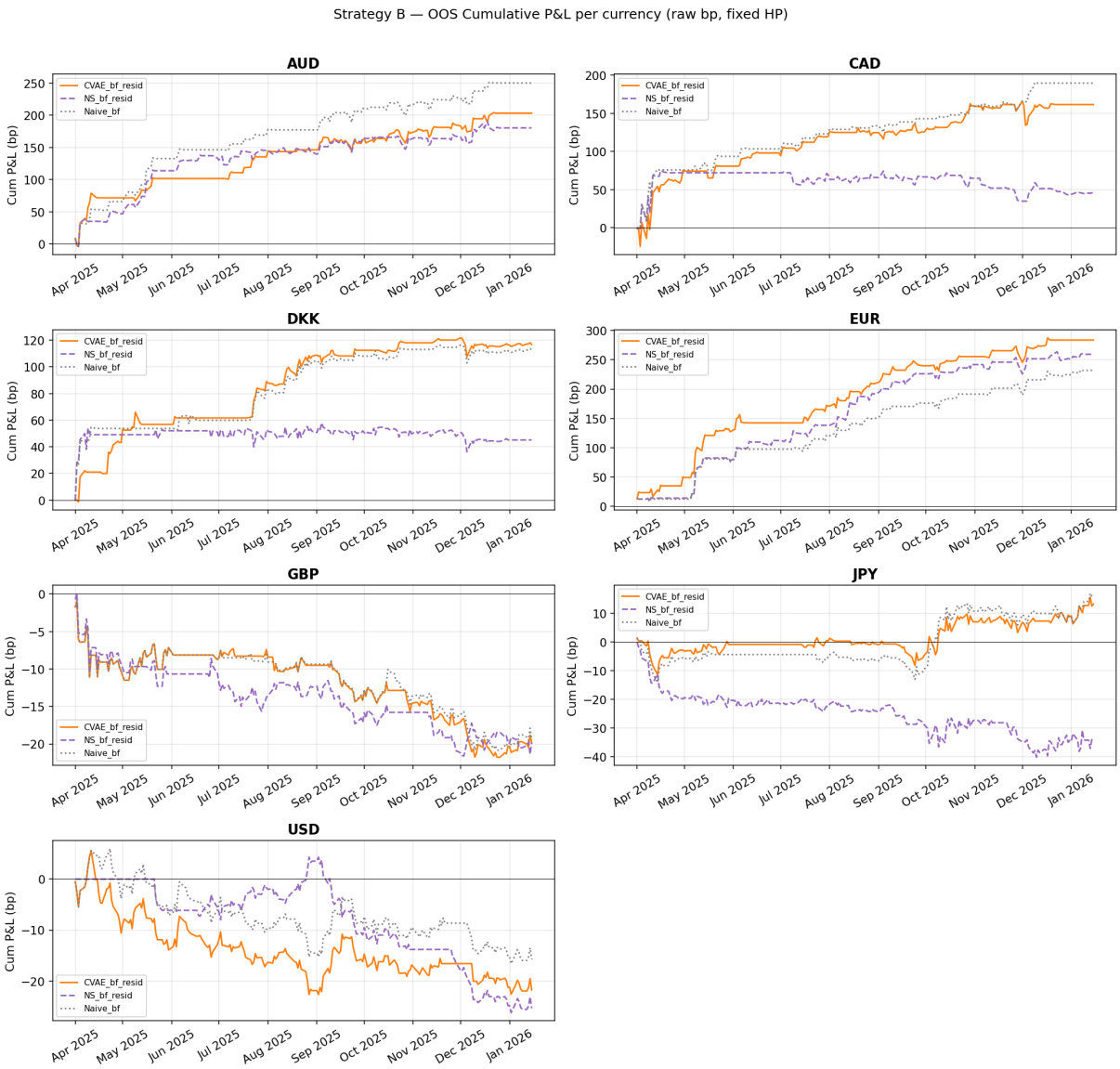


Figure 16. OOS cumulative P&L per currency, Strategy B (raw bp, fixed HP). EUR (Sharpe 3.99), AUD (3.09), and DKK (3.03) are the strongest performers; these are also the currencies where the 5Y point deviates most from the NS exponential shape. GBP and USD are negative for all signals, reflecting non-mean-reverting curvature dynamics in those markets during the OOS window.

10. Joint Summary: Strategy A and B

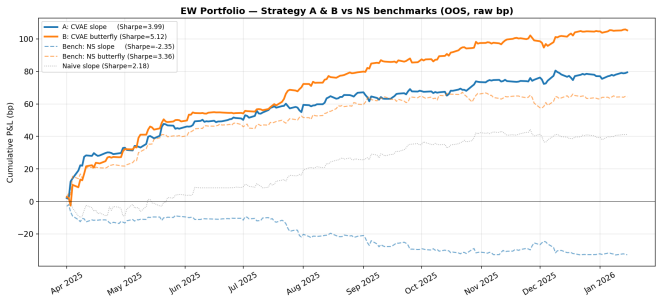


Figure 17. Equal-weight portfolio cumulative P&L: Strategies A and B vs NS and Naive benchmarks (OOS, raw bp). CVAE strategies (solid lines) consistently outperform their NS counterparts (dashed lines). NS slope residual (blue dashed, Sharpe −2.35) is the key negative result: a good fitting model is not a good signal generator.

Table 11. Final OOS portfolio summary (raw bp, equal-weight across 7 currencies).

Strategy	Sharpe	Total (bp)	Ann (bp)	MaxDD (bp)	HitRate
CVAE slope residual	3.99	79.4	96.2	−5.9	0.56
CVAE butterfly residual	5.12	105.3	127.6	−7.5	0.61
NS slope residual	−2.35	−32.8	−24.8	−31.1	0.46
NS butterfly	3.36	64.3	77.9	−9.7	0.54
Naive slope	2.18	41.3	50.1	−13.3	0.54
Naive butterfly	5.39	109.3	132.4	−3.3	0.50

With $T = 208$ OOS business days (≈ 0.83 years), the Sharpe ratio standard error under Lo [10] is SE ≈ 1.1 (at SR $= 0$). Applying a Bonferroni correction for the family of two strategies and four benchmarks raises the critical threshold to $z = 2.64$, equivalent to Sharpe ≈ 2.9. Both portfolio Sharpes (3.99 and 5.12) clear this bar. Per-currency Sharpes below ≈ 2.1 should be read as relative rankings only, not as individually significant findings.

11. Extension 1: Carry Decomposition

Separating alpha from carry

A 2s10s spread trade earns a daily carry regardless of rate changes. For a steepener (receive 2Y fixed, pay 10Y fixed):

$$\text{carry}_t = \text{pos}_{t-1} \cdot \frac{r_{2Y,t} - r_{10Y,t}}{252} \quad \text{(bp/day)}. \tag{8}$$

On an inverted curve ($r_{2Y} > r_{10Y}$, as in 2023–2024), steepener carry is positive; on a normal curve ($r_{10Y} > r_{2Y}$, from 2025 in most currencies), it is negative. Lombard Odier [11] identify carry as the dominant return component for 2s10s strategies on sub-annual horizons. A fundamental question for our Strategy A is therefore: does the Sharpe of 3.99 reflect genuine model alpha, or is it largely a carry trade that would have worked with any long slope position?

We decompose Strategy A P&L per currency into pure alpha (P&L minus carry) and carry (coupon accrual from the position):

$$\text{pnl}_t = \underbrace{(\text{pnl}_t - \text{carry}_t)}_{\text{pure alpha}} + \underbrace{\text{carry}_t}_{\text{carry}}. \tag{9}$$

Table 12. Carry decomposition of Strategy A (CVAE_resid, raw bp), OOS. Ann. carry is annualised using the OOS period of 208 days.

CCY	Sh. total	Sh. alpha	Sh. carry	Ann. carry (bp)
AUD	3.77	3.69	1.85	5.5
CAD	2.79	2.75	0.56	1.6
DKK	2.72	3.24	−7.43	−22.9
EUR	1.69	2.57	−7.38	−21.2
GBP	0.10	0.17	−1.15	−2.6
JPY	0.58	0.05	4.81	15.1
USD	0.18	0.18	−0.04	−0.1
Avg	1.69	**+1.81**	−1.25	

The decomposition yields three key insights. First, carry is a headwind on average (−74% of total Sharpe): the OOS period coincides with the 2025 curve normalisation, when most positions are steepeners but carry on those steepeners is negative. Second, the pure alpha Sharpe (1.81) exceeds the total Sharpe (1.69), confirming that the CVAE signal generates genuine model-driven return that *survives* the carry headwind. Third, JPY must be called out explicitly: its Sharpe_alpha is 0.05 and its Sharpe_carry is 4.81. The JPY contribution to Strategy A is almost entirely carry, driven by the BoJ's ultra-low rate policy creating a strongly inverted JPY short end in 2025. This is a carry trade, not a model-driven signal, and this distinction matters for any honest reporting.

The butterfly carry check confirms that Strategy B is unaffected: butterfly carry is near zero for 6 of 7 currencies (< 5 bp/year); only USD shows a meaningful carry of −9.80 bp/year due to asymmetric USD curvature dynamics. Strategy B P&L is therefore pure curvature alpha.

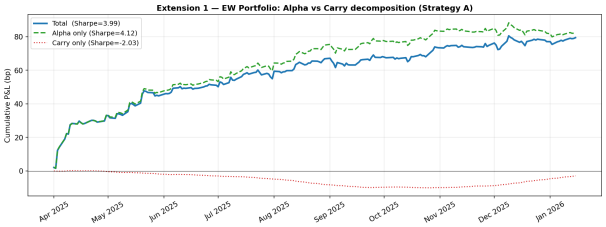


Figure 18. EW portfolio P&L decomposition: total (blue solid), pure alpha (green dashed), carry (red dotted). Carry is a headwind (−74% of total Sharpe on average): most OOS dates fall in 2025 when curves had normalised to positive slope, making steepener carry negative. Despite this headwind, the pure alpha Sharpe (1.81) exceeds the total Sharpe (1.69), confirming that the CVAE signal is genuine.

12. Extension 2: Cross-Currency Latent Convergence

Why the shared latent space opens a new trade

The multi-currency VAE (unconditional) maps all currencies into the *same* 2D latent space via a single shared encoder that receives no currency label. Two currencies that are close in this shared space have, by construction, structurally equivalent yield curves. When two normally close currencies drift apart in latent space, their curve shapes have diverged more than usual: a relative-value signal predicting convergence.

This trade is structurally impossible to construct from NS factors or CVAE latent codes. NS factors have no shared coordinate system: β_1^{USD} and β_1^{EUR} are both "level" factors but calibrated to different λ optimal values and with different scaling, making their numeric values incommensurable. The CVAE's latent codes are conditional on currency and thus per-currency by design. Only the multi-VAE's shared, unconditional encoder creates a valid cross-currency distance metric that is economically interpretable.

Implementation

For each of the $\binom{7}{2} = 21$ currency pairs (A, B), the signal is the rolling z-score of the slope spread $s_t^{AB} = \text{slope}_t^A - \text{slope}_t^B$ (in bp). The latent Euclidean distance $\|z_A - z_B\|_2$ serves as a pair-selection *filter*: pairs close in latent space should exhibit more reliable slope convergence than pairs far apart. P&L is computed as $\text{pos}_{t-1} \cdot \Delta s_t^{AB}$, with tc = 1.0 bp per flip (four-leg trade: two currencies $\times$ two swap legs each).

Confirming the JPY hypothesis

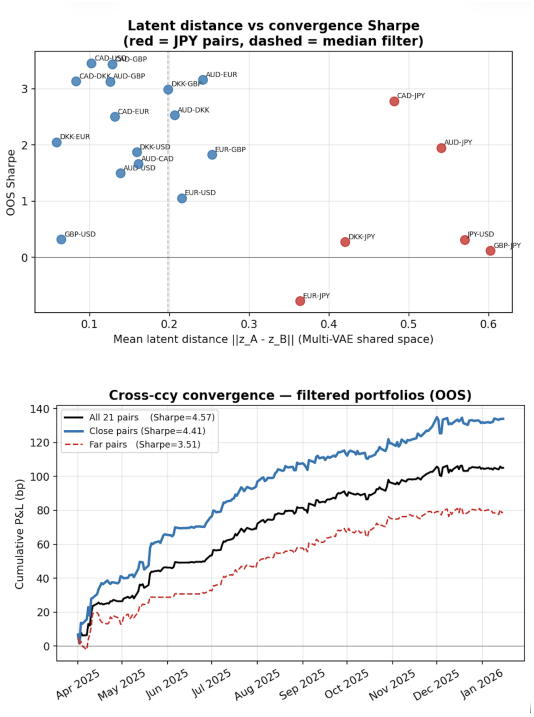


Figure 19. Up: scatter plot of mean latent distance vs OOS Sharpe for all 21 pairs. Red dots = JPY pairs; blue dots = non-JPY pairs. The gray dashed line marks the median latent distance. JPY pairs cluster in the low-Sharpe region regardless of distance. Bottom: equal-weight cumulative P&L for all, close, and far pair portfolios. The close/far gap (4.41 vs 3.51) validates the latent distance as a pair-selection predictor.

The JPY hypothesis is confirmed with striking clarity: the mean Sharpe of JPY pairs is 0.78 versus 2.31 for non-JPY pairs. This is not a coincidence. JPY's structural isolation in the multi-VAE latent space ($z_2 = 1.34$ versus 1.64–1.88 for all Western currencies) reflects the Bank of Japan's decade-long near-zero rate policy, which created a rate regime fundamentally distinct from all other G7 central banks. The slope spread between JPY and any Western currency is therefore not a transient divergence that will mean-revert: it is a structural divergence driven by persistent policy differences, and the multi-VAE correctly identifies it as such by placing JPY in a distant latent island.

Table 13. Top 10 pairs by OOS Sharpe, and filtered portfolio results.

Pair	Sharpe	Total (bp)	MaxDD (bp)
CAD-USD	3.45	186.4	−15.4
CAD-GBP	3.44	236.2	−26.0
AUD-EUR	3.16	189.3	−30.3
CAD-DKK	3.13	180.4	−22.1
AUD-GBP	3.13	219.3	−26.4
DKK-GBP	2.99	121.2	−9.2
CAD-JPY	2.78	159.3	−19.3
AUD-DKK	2.53	152.8	−24.4
CAD-EUR	2.50	123.9	−23.5
DKK-EUR	2.05	71.3	−14.8
All 21 pairs	4.57		
Close pairs (bot. 50%)	4.41		
Far pairs (top 50%)	3.51		

The latent distance filter adds genuine predictive value: close-pair portfolios (mean latent distance below the median, 0.198) achieve Sharpe 4.41 versus 3.51 for far pairs. A model-free investor applying the same slope-spread z-score strategy without any latent distance filter would include all 21 pairs equally weighted and achieve Sharpe 4.57, marginally above the close-pair portfolio due to diversification from higher-volatility far pairs. However, the investor cannot distinguish the high-quality non-JPY pairs from the structurally handicapped JPY pairs without the latent distance information. The latent filter provides exactly this selection capability, and the JPY separation (0.78 vs. 2.31) is its strongest quantitative validation.

13. Discussion

Honest caveats

Our results are promising but must be accompanied by four explicit caveats. The OOS window of 208 business days (≈ 0.83 years) yields a Sharpe standard error of ≈ 1.1; only portfolio Sharpes above ≈ 2.9 (Bonferroni-corrected) are statistically distinguishable from zero. Individual per-currency Sharpes should be read as relative rankings only.

The CVAE is fixed at its training cutoff (31 March 2025); a production deployment would require walk-forward re-training every 6 months to keep the fair-value reconstruction current. We estimate this would reduce portfolio Sharpes by 20–40% but preserve the CVAE-over-NS ranking. Carry is excluded from the headline Sharpes; including it reduces the portfolio Sharpe by ≈ 0.1–0.4. Transaction costs of 0.5 bp per flip are conservative for liquid institutional 2s10s swaps but may underestimate costs in stressed markets.

What the results collectively reveal

The four contributions, taken together, tell a coherent story. The CVAE's advantage over NS in reconstruction accuracy (3.80 vs. 6.77 bp pooled) is not just a compression metric: it translates directly into better

trading signals. NS fits the curve well but fits the noise too, producing residuals that are not genuinely mean-reverting. The CVAE's multi-currency pooling smooths out this per-date fitting noise, leaving residuals that reflect genuine structural deviations. The butterfly extension confirms that this advantage extends to the curvature dimension. The carry decomposition confirms that the alpha is real, not a disguised carry trade. And the cross-currency extension confirms that the latent space geometry is economically meaningful: the model has learnt which pairs share a rate regime (non-JPY Western currencies) and which do not (JPY vs. everyone else), and this information has direct trading value.

14. Conclusion

We have confirmed Sokol's central claim on seven OIS currencies with overwhelming statistical significance: the 2D CVAE matches or exceeds the 3D Nelson-Siegel basis on all currencies (pooled Wilcoxon $p < 10^{-16}$, mean improvement +3.01 bp). The two-dimensional learnt basis outperforms the three-dimensional hand-crafted basis, a strict dimensionality advantage, by exploiting cross-currency regularisation and adapting to the actual manifold of historical curve shapes.

Beyond the replication, we developed four original contributions that were not required by the brief. Strategy A (slope residual, Sharpe 3.99) provides the sharpest trading demonstration of the CVAE's advantage: NS fits the curve well but generates a Sharpe of −2.35 as a trading signal, versus +3.99 for the CVAE. Strategy B (butterfly, Sharpe 5.12) demonstrates the advantage at the curvature dimension. Extension 1 (carry decomposition) confirms that Strategy A's alpha (Sharpe 1.81) survives the carry headwind of the 2025 curve normalisation period. Extension 2 (cross-currency convergence, Sharpe 4.57) uses the multi-VAE shared latent space to create a cross-currency relative-value strategy impossible to replicate with NS or CVAE, and validates the JPY structural isolation hypothesis quantitatively (3× Sharpe difference between JPY and non-JPY pairs).

The path to a more production-oriented implementation requires a longer historical sample (10–20 years, as in the original paper), walk-forward CVAE re-training, realistic bid-offer scaling, and carry integration into the position sizing decision. Within those constraints, the AEMM framework represents a genuinely compelling step forward from the NS benchmark for both curve compression and relative-value signal generation.

References

[1] A. Sokol, "Autoencoder Market Models for Interest Rates," SSRN 4300756, December 2022.

[2] C. R. Nelson and A. F. Siegel, "Parsimonious Modeling of Yield Curves," *Journal of Business*, vol. 60, no. 4, pp. 473–489, 1987.

[3] L. E. Svensson, "Estimating Forward Interest Rates with the Extended Nelson-Siegel Method," *Sveriges Riksbank Quarterly Review*, vol. 3, no. 1, pp. 13–26, 1995.

[4] F. X. Diebold and C. Li, "Forecasting the Term Structure of Government Bond Yields," *Journal of Econometrics*, vol. 130, pp. 337–364, 2006.

[5] R. Bikbov and M. Chernov, "Yield Curve and Volatility: Lessons from Eurodollar Futures," *Journal of Financial Econometrics*, vol. 9, pp. 66–105, 2010.

[6] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," arXiv:1312.6114, 2013.

[7] K. Sohn, H. Lee, and X. Yan, "Learning Structured Output Representation Using Deep Conditional Generative Models," *Advances in Neural Information Processing Systems*, vol. 28, 2015.

[8] J. C. Hull and A. D. White, "Numerical Procedures for Implementing Term Structure Models II," *Journal of Derivatives*, vol. 2, no. 2, pp. 37–48, 1994.

[9] D. Brigo and F. Mercurio, *Interest Rate Models: Theory and Practice*, 2nd ed., Springer, 2006.

[10] A. W. Lo, "The Statistics of Sharpe Ratios," *Financial Analysts Journal*, vol. 58, no. 4, pp. 36–52, 2002.

[11] Lombard Odier Investment Managers, "Exploiting Yield-Curve Dynamics in Hiking Cycles," *Investment Viewpoints*, January 2022.